

HE2003 Econometrics I

Tutorial 6

23/09/2024

Question 1

Explain the notions of hypothesis testing, critical value, null distribution, alternative distribution, acceptance region, and rejection region. Use specific examples if necessary.

Hypothesis testing is a statistical method which entails formulating a specific hypothesis about the population and then using sample evidence to decide whether to accept or reject the hypothesis. We take an example of hypothesis tests concerning the population mean (Does μ , the population mean of Y , equal a specific value, μ_0). The first step is to specify the hypothesis to be tested, called the null hypothesis:

$$H_0: \mu = \mu_0$$

The alternative hypothesis specifies what is true if the null hypothesis is false. A two-sided alternative hypothesis is that μ is different from μ_0 :

$$H_1: \mu \neq \mu_0 \text{ (two-sided alternative)}$$

Question 1

To perform the hypothesis test, a test statistics is required. In our example, we use the t-statistic:

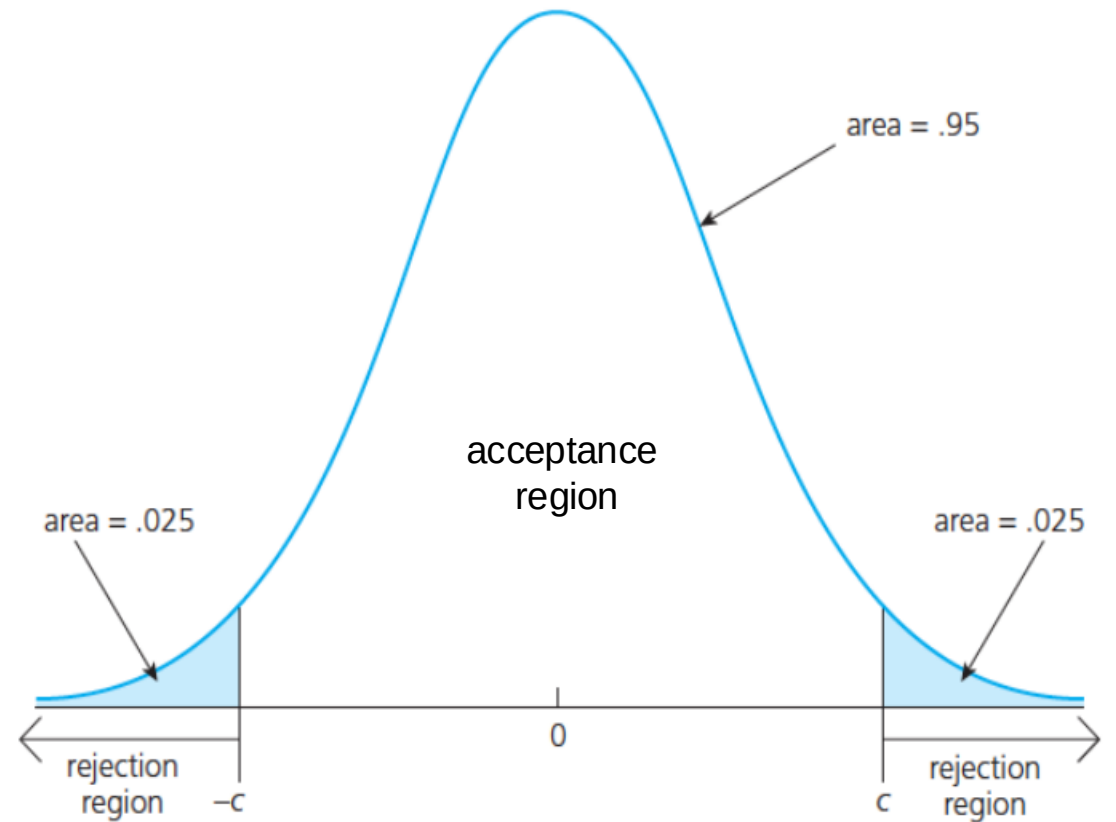
$$t = \frac{\bar{Y} - \mu_0}{SE(\bar{Y})}$$

where $\bar{Y}$ is sample average and $SE(\bar{Y})$ is the standard error of $\bar{Y}$.

The null distribution represents the distribution of the test statistic if the null hypothesis is true, while the alternative distribution reflects the distribution of the test statistic when the alternative hypothesis is true. In this case, if the null hypothesis is true, the t-statistic follows $N(0,1)$, a standard normal distribution, in large samples (we focus on the large-sample distribution).

Question 1

The critical value of the test statistic is the threshold at which the null hypothesis is just rejected at a specified significance level, such as $\alpha = 0.05$. The rejection region is the range of values for the test statistic that lead to rejecting the null hypothesis, while the acceptance region includes the range of values for which the null hypothesis is not rejected. The diagram shows the distribution of the test statistic under null hypothesis, along with the acceptance and rejection regions for a two-sided test at $\alpha = 0.05$, and c is the critical value.



Question 2

Consider the linear regression model: $Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n$

(a) Explain the distribution of the OLS estimator of β_1 in large samples.

When the OLS assumptions are satisfied, $\hat{\beta}_1$ is an unbiased estimator of the true value β_1 . In large samples, $\hat{\beta}_1$ follows a normal distribution centered at β_1 :

$$\hat{\beta}_1 \sim N\left(\beta_1, \sigma_{\hat{\beta}_1}^2\right),$$

where $\sigma_{\hat{\beta}_1}^2$ denotes the variance of $\hat{\beta}_1$ under large samples.

Note: In this course, we always focus on the large-sample distributions (e.g., the normal distribution) instead of small-sample distributions.

Question 2

(b) Suppose we want to test the hypothesis $H_0: \beta_1 = 1$ vs. $H_0: \beta_1 \neq 1$. Write down the formula of the t-statistic. Explain how we can obtain the large-sample distribution of this t-statistic using the result in (a).

To test the hypothesis $H_0: \beta_1 = 1$, the corresponding t-test statistic is:

$$t = \frac{\hat{\beta}_1 - 1}{SE(\hat{\beta}_1)}$$

Given $\hat{\beta}_1 \sim N(\beta_1, \sigma_{\hat{\beta}_1}^2)$, we can do the following transformation (standardization) for $\hat{\beta}_1$:

$$\frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \sim N(0,1)$$

where $\sigma_{\hat{\beta}_1}$ denotes the standard deviation of $\hat{\beta}_1$, and equals to the square root of the variance $\sigma_{\hat{\beta}_1}^2$

Question 2

We have the following decomposition:

$$t = \frac{\hat{\beta}_1 - 1}{SE(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} + \frac{\beta_1 - 1}{SE(\hat{\beta}_1)}$$

When H_0 is true, $\beta_1 = 1$ and $\frac{\beta_1 - 1}{SE(\hat{\beta}_1)} = 0$. Therefore, when H_0 is true:

$$t = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)}$$

Furthermore, we note that the standard error $SE(\hat{\beta}_1)$ is the estimator of $\sigma_{\hat{\beta}_1}$, so under large samples, $SE(\hat{\beta}_1)$ will be very close to $\sigma_{\hat{\beta}_1}$. Thus, we can replace $\sigma_{\hat{\beta}_1}$ with $SE(\hat{\beta}_1)$. Hence, when the OLS assumptions are satisfied and $H_0: \beta_1 = 1$ is true, we can conclude that the t-statistic also follows $N(0, 1)$ under large samples.

$$t = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \sim \frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \sim N(0, 1)$$

Question 3

Suppose that a random sample of size $n = 1,000$ is drawn from certain population and the observations (Y_i, X_i) , $i = 1, \dots, 1,000$ satisfy the OLS assumptions. The result of the OLS regression is:

$$\hat{Y}_i = 1.5 + 2.4X_i, \quad i = 1, \dots, 1000,$$

with the standard error of $\hat{\beta}_0$ equal to 1.0 and the standard error of $\hat{\beta}_1$ equal to 1.2.

(a) Test the slope of the regression using $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$ at the 5% significance level. Similarly, test the intercept of the regression at the 5% significance level.

(b) Suppose that X_i actually has no effect on Y_i . Also suppose that 2,000 random samples of size $n = 1,000$ are drawn repeatedly from the population, and the OLS regressions are estimated repeatedly 2,000 times (suppose the sampling process always satisfies the OLS assumptions). In how many of these random samples would the null hypothesis $H_0: \beta_1 = 0$ be rejected by the (two-sided) t-test at the 10% significance level? Explain.

Question 3

(a) Test the slope and intercept at 5% significance level

The (two-sided) critical value at the 5% significance level is 1.96.

Test $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$:

$$|t_{\beta_1}| = \left| \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \right| = \left| \frac{2.4 - 0}{1.2} \right| = 2 > 1.96$$

$\Rightarrow$ the zero null hypothesis for the slope is **rejected** at the 5% level.

Test $H_0: \beta_0 = 0$ vs $H_1: \beta_0 \neq 0$:

$$|t_{\beta_0}| = \left| \frac{\hat{\beta}_0 - 0}{SE(\hat{\beta}_0)} \right| = \left| \frac{1.5 - 0}{1.0} \right| = 1.5 < 1.96$$

$\Rightarrow$ the zero null hypothesis for the intercept is **not rejected** at the 5% level.

Note: $n = 1,000$ is considered a large sample.

Question 3

(b) In how many of these random samples would the null hypothesis $H_0: \beta_1 = 0$ be rejected

As X actually has no effect on Y , the null hypothesis $H_0: \beta_1 = 0$ is true. Based on the definition of the 10% significance level (tolerance level for **Type I error**), the t-test has **Type I error** equal to 10%: even when H_0 is true, the t-test still can reject H_0 10% of the time: the probability of rejecting H_0 even when H_0 is true is equal to 10% (that is, $P_{H_0}(\text{Reject } H_0) = 10\%$).

Now since 2,000 random samples are repeatedly drawn from the population, we expect that among these samples the t-test will reject $2,000 \times 10\% = 200$ of them.

		Truth (for population)	
Null hypothesis H_0		True	False
Decision based on sample	Rejected	Type I Error False positive (Prob = α)	Correct decision True positive
	Not Rejected	Correct decision True negative	Type II Error False negative

Question 4

(a) Consider the following linear regression model for saving and income:

$$savings_i = \beta_0 + \beta_1 income_i + u_i, \quad u_i = \sqrt{income_i} \times e_i$$

where e_i satisfies $E[e_i] = 0$ and $Var[e_i] = \sigma^2$. In addition, e_i is independent of $income_i$.

First, show that the conditional expectation of e_i given the value of $income_i$ is equal to zero (by using the property of independence between two random variables). Second, show that this model satisfies the OLS Assumption 1. [Hint: Notice that the following result holds for random variables X and Y for any function $g(X)$: $E(g(X) \times Y|X) = g(X) \times E(Y|X)$.]

(b) Consider the linear regression model : $Y_i = \beta_0 + \beta_1 X_i + u_i$. Suppose you know that $\beta_0 = 0$. Derive a closed-form formula for the OLS estimator of β_1 . Hint: As we know $\beta_0 = 0$, there is no need to estimate it in this case. Therefore, the minimization problem becomes:

$$\min_{b_1} \sum_{i=1}^n (Y_i - 0 - b_1 X_i)^2, \text{ i.e. } \min_{b_1} \sum_{i=1}^n (Y_i - b_1 X_i)^2$$

Question 4

(a) Proof this model satisfies the OLS Assumption 1

Firstly, due to the independence between e_i and $income_i$:

$$E[e_i|income_i] = E[e_i] = 0$$

Secondly, we have:

$$E[u_i|income_i] = E[(\sqrt{income_i} \times e_i)|income_i]$$

From the property of conditional expectation:

$$E[u_i|income_i] = \sqrt{income_i} \times E[e_i|income_i] = 0$$

Thus, the OLS Assumption 1 is satisfied.

Question 4

(b) The minimization problem: $\min_{b_1} \sum_{i=1}^n (Y_i - b_1 X_i)^2$

Take the first derivatives with respect to b_1 :

$$\frac{d}{db_1} \sum_{i=1}^n (Y_i - b_1 X_i)^2 = -2 \sum_{i=1}^n (Y_i - b_1 X_i) X_i$$

The OLS estimator of the slope/gradient $\hat{\beta}_1$ satisfies the FOC:

$$\begin{aligned} \sum_{i=1}^n (Y_i - \hat{\beta}_1 X_i) X_i &= \sum_{i=1}^n X_i Y_i - \sum_{i=1}^n \hat{\beta}_1 X_i^2 = 0 \\ \Rightarrow \hat{\beta}_1 \sum_{i=1}^n X_i^2 &= \sum_{i=1}^n X_i Y_i \Rightarrow \hat{\beta}_1 = \frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2} \end{aligned}$$